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Next item →

1. A cardiac monitor is used to measure the heart rate of a patient after surgery. It compiles the number of heartbeats after t minutes. The data collected from the monitor are shown in the table below. 1 / 1 point

t (minutes)	Heartbeats
26	2530
28	2661
30	2806
32	2948
34	3080

When the data in the table are graphed, the slope of the tangent line represents the heart rate in beats per minute. The monitor estimates this value by calculating the slope of a secant line.

Find the slope of the secant line between the points where $t = 26$ and $t = 32$.

If your answer is not a whole number, enter it as a fraction in lowest terms.

$\frac{209}{3}$

209/3

Nice work

The slope of the secant line is given by $\frac{\text{change in heartbeats}}{\text{time elapsed}}$.

Between $t = 26$ and $t = 32$, the change in heartbeats is $2948 - 2530 = 418$ and the time elapsed is $32 - 26 = 6$.

Thus the slope of the secant line between these points is $\frac{418}{6} = \frac{209}{3}$.

2. Referring to the table in the previous question, compute the slope of the secant line between the points with the given values of t: 1 / 1 point

- $t = 28$ and $t = 32$
- $t = 30$ and $t = 32$
- $t = 32$ and $t = 34$

Use the data to estimate the patient's heart rate after 32 minutes.

Round to the nearest integer

68

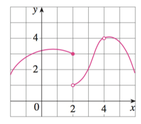
Nice work

Compute the slopes of the secant lines with the same method from Question 1.

Comparing the values of the slopes, we can see that the patient's heart rate at $t = 32$ is somewhere between 66 and 70 beats per minute.

3. The function $y = f(x)$ is graphed below. Use the graph to find the values, if they exist, of $\lim_{x \to 2^-} f(x)$, $\lim_{x \to 2^+} f(x)$, $\lim_{x \to 2} f(x)$, and $f(2)$. 1 / 1 point

Select one answer for each.



- [ ] $\lim_{x \to 2^-} f(x) = 1$
- [x] $\lim_{x \to 2^-} f(x) = 3$

Nice work

This is the left-hand limit of $f(x)$ as x approaches 2. To find the limit, look at the values of $f(x)$ for x-values that are less than 2.

- [ ] $\lim_{x \to 2^+} f(x) = 3$
- [ ] $\lim_{x \to 2^+} f(x)$ does not exist.
- [ ] $\lim_{x \to 2} f(x) = 1$
- [ ] $\lim_{x \to 2} f(x) = 3$
- [x] $\lim_{x \to 2} f(x)$ does not exist.

Nice work

The limit exists if and only if both one-sided limits exist and are equal to the same number.

- [ ] $f(2) = 1$
- [x] $f(2) = 3$

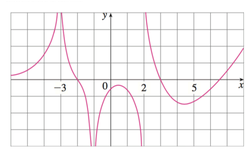
Nice work

The function value can be read from the graph, independently of the limits.

- [ ] $f(2)$ does not exist.

4. $y = A(x)$ is graphed below. Find the following limits: $\lim_{x \to -3} A(x)$, $\lim_{x \to -1} A(x)$, $\lim_{x \to 2^-} A(x)$, and $\lim_{x \to 2^+} A(x)$. 1 / 1 point

Select one answer for each limit.



- [ ] $\lim_{x \to -3} A(x) = -\infty$
- [x] $\lim_{x \to -3} A(x) = +\infty$

Nice work

If the function values become arbitrarily large positive numbers close to the point, then the limit is equal to $+\infty$.

- [x] $\lim_{x \to -1} A(x) = -\infty$

Nice work

If the function values become arbitrarily large negative numbers close to the point, then the limit is equal to $-\infty$.

- [ ] $\lim_{x \to -1} A(x) = +\infty$
- [x] $\lim_{x \to 2^-} A(x) = -\infty$

Nice work

If the function values become arbitrarily large negative numbers close to the point, then the limit is equal to $-\infty$.

Remember that this limit is the left-hand limit, which means we look at x-values that are less than 2.

- [ ] $\lim_{x \to 2^-} A(x) = +\infty$
- [ ] $\lim_{x \to 2^-} A(x) = -\infty$
- [x] $\lim_{x \to 2^+} A(x) = +\infty$

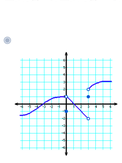
Nice work

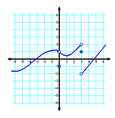
If the function values become arbitrarily large positive numbers close to the point, then the limit is equal to $+\infty$.

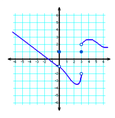
Remember that this limit is the right-hand limit, which means we look at x-values that are greater than 2.

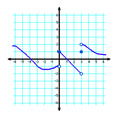
5. Choose the graph of a function f that satisfies all the given conditions. 1 / 1 point

$\lim_{x \to 0} f(x) = 1$, $\lim_{x \to -2^-} f(x) = -2$, $\lim_{x \to 2^+} f(x) = 2$, $f(0) = -1$, $f(3) = 1$









Nice work

Because $f(0) = -1$ and $f(3) = 1$, the points $(0, -1)$ and $(3, 1)$ are in the graph of f.

$\lim_{x \to 0} f(x) = 1$ means the values of f approach 1 as x approaches 0 from either side.

$\lim_{x \to -2^-} f(x) = -2$ means f approaches -2 as x approaches -3 from the left. $\lim_{x \to 2^+} f(x) = 2$ means f approaches 2 as x approaches 3 from the right.

6. Select all of the vertical asymptotes of $f(x) = \frac{x^2 - x - 2}{x^2 + x - 2}$ 1 / 1 point

- [x] $x = 1$

Nice work

To find vertical asymptotes, find those points where the denominator equals zero and the limit of the function is undefined.

The equation $x^2 + x - 2 = 0$ can solved by factoring or with the quadratic formula.

- [ ] $x = 2$
- [ ] $f(x)$ has no vertical asymptotes.
- [ ] $x = -1$
- [x] $x = -2$

Nice work

To find vertical asymptotes, find those points where the denominator equals zero and the limit of the function is undefined.

The equation $x^2 + x - 2 = 0$ can solved by factoring or with the quadratic formula.

7. Evaluate the following as true or false: 1 / 1 point

The notation $\lim_{x \to 2} f(x) = 5$ states that the limit of the function f at $x = 5$ is 2.

- ( ) True
- (•) False

Nice work

The notation states that the limit of the function f at $x = 2$ is 5.

8. $p(t) = \begin{cases} t + 2, & t < -2 \\ t^3, & -2 \le t \le 2 \\ t + 4, & t > 2 \end{cases}$. 1 / 1 point

Evaluate $\lim_{t \to -2} p(t)$.

8

Nice work

The negative sign in the limit indicates a left-handed limit. For left-handed limits, you only have to consider the domain values less than the one being approached in the limit.

For t-values that are close to 2 and less than 2, $p(t) = t^3$.

Thus $\lim_{t \to -2} p(t) = \lim_{t \to -2} t^3 = 2^3 = 8$

9. Evaluate $\lim_{x \to 2} \sqrt{x^2 - 4}$. 1 / 1 point

- (•) The limit is undefined
- ( ) -2
- ( ) 0
- ( ) 2

Nice work

This is a left hand limit, so we consider values that are close to 2 and less than 2. If $x < 2$, then $x^2 < 4$, so the expression inside the square root is negative.

10. Determine $\lim_{x \to 0} |x|$. 1 / 1 point

- ( ) -1
- ( ) The limit is undefined
- (•) 0
- ( ) 1

Nice work

Recall that the absolute value function is a piecewise function,

$|x| = \begin{cases} -x, & x \le 0 \\ x, & x > 0 \end{cases}$

To find the limit as $x \to 0$, we evaluate both the left- and right-hand limits at $x = 0$ using the piecewise definition.